Predicting Yield Curve Inversions using Machine Learning Methods

ECO482 Research Project Report

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Introduction

This report focuses on applying machine learning techniques to predict yield curve inversions. The yield curve is an important tool in economics, used to forecast interest rates, economic growth, and, most importantly, recessions. An inverted yield curve occurs when short-term debt instruments pay a higher yield than long-term debt instruments (TD Bank, 2025), and this typically reflects a shift in investor preferences toward longer-duration bonds. As demand for long-term bonds rises, their yields decline, while short-term yields rise to remain competitive and attract capital. This pattern can emerge for several reasons, but it is commonly interpreted as a signal of weakening economic expectations, with investors anticipating that central banks may cut interest rates to encourage borrowing and support economic activity. The yield curve has inverted before every U.S. recession in the past 60 years (Benzoni, 2018), making it a highly reliable recession indicator. Predicting these inversions in advance can give policymakers a head start in setting up economic safeguards before a recession hits. Our report addresses three important questions. First, can machine learning help detect recession signals earlier and more accurately? Second, which macroeconomic factors actually drive these predictions? Finally, what are the possible limitations of these models, and how can they be addressed properly? This report extends the ideas of Rayeni and Naderi (2025) to the U.S. economy.

Methodology

To construct the dependent variable, we first calculate the yield curve slope as the difference between the 10-year and 3-month Treasury yields, which serves as the primary signal underlying our predictive model. We define an indicator variable I_t as follows:

$$I_t = \begin{cases} 1 & \text{if slope } < 0 \\ 0 & \text{if slope } \geq 0 \end{cases}$$

We then apply a 6-month forward shift to this binary variable to construct our target variable, Y_t, which represents whether a yield curve inversion occurs six months into the future. Finally, we apply the transformation codes outlined in our data appendix to each independent variable in our dataset, ensuring that we work with changes over time rather than raw levels. (For a detailed account of our cleaning and transformation procedure, please see the accompanying code on my GitHub.)

Data and Descriptive Statistics

We use the FRED-MD Monthly Database, sourced from the St. Louis Federal Reserve for our analysis. This dataset is readily available online and is commonly used among economists for macroeconomic forecasting purposes. It consists of monthly variables between January 1963 and December 2025, with a total of 804 monthly observations. It includes 123 macroeconomic variables across 8 categories, such as Output, Labor, Housing, Money, Consumption, Interest Rates, Stock Market and Prices. Due to the large number of predictors used in our models, we could not report the summary statistics for each predictor. However, below is a table consisting of some important variables in the dataset.

Table 1: Summary Statistics of Variables

Variable	**Description**	**Mean**	**Std Dev**	**Min**	**Max**
GS10	10Y Treasury Yield	5.75	2.90	0.62	15.32
TB3MS	3M Treasury Yield	4.36	3.10	0.01	16.30
Slope	GS10 - TB3MS	1.43	1.27	-2.65	4.42
UNRATE	Unemployment Rate (%)	5.86	1.68	3.40	14.80
S&P500	S&P 500 Composite Index	1,034.9	1,355.56	53.73	6,853.03

Figure 1 displays a graph of the slope of the yield curve over time. We see inversions around the 2008 stock market crash, the dot com bubble, and most notably around the late 1970's

and early 1980's. Most notably, a yield curve inversion is not a common event, which can make our task more challenging.

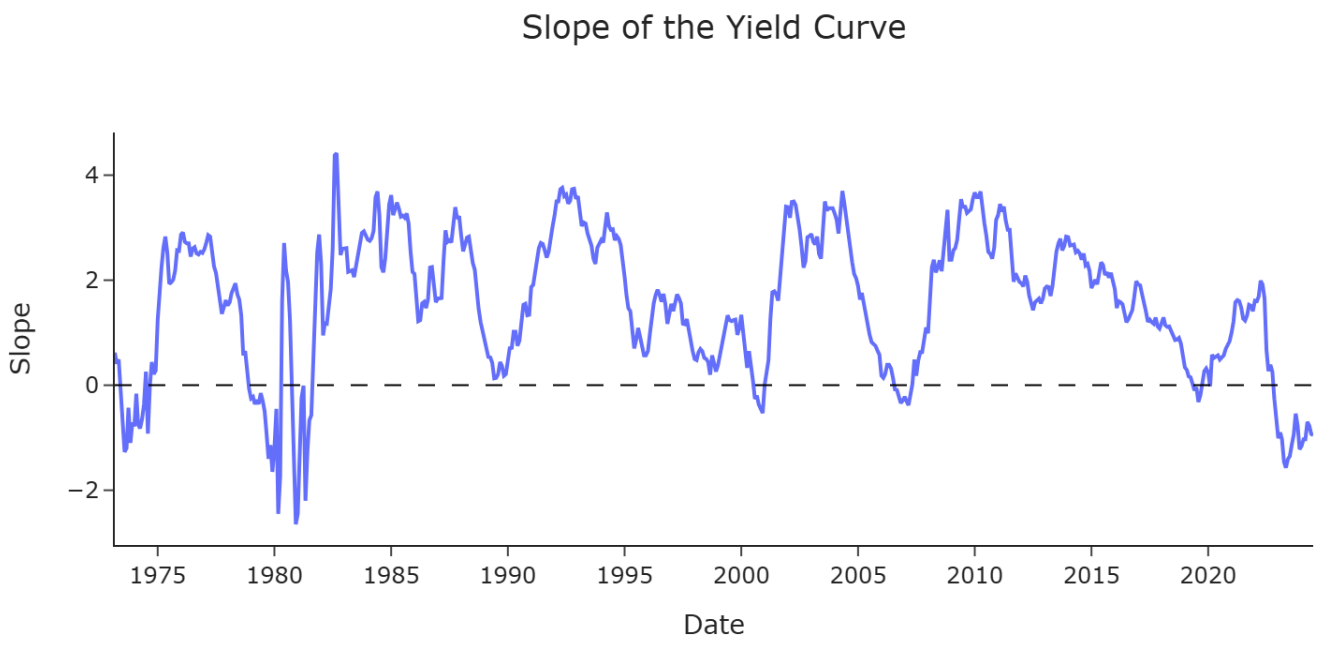


Figure 1: Slope of the Yield Curve from March 1973 to Sept 2025

Models

We use a random walk as our baseline model, alongside a Random Forest and an XGBoost Classifier. Model performance is optimized and evaluated using several methods, including Precision-Recall curves, ROC curves, and confusion matrices. To prevent look-ahead bias, we implement a Time Series Split with 2 splits. We use the $F1$ score as our primary evaluation and tuning metric, as it is particularly well-suited to imbalanced data.

Random Walk

We use a random walk as a baseline. Simply put, a random walk assumes that the yield curve's inversion status 6 months from now will be the same as it is today. We define it mathematically as

$$\hat{Y}_t = I_{t-6}$$

where I_{t-6} is the observed inversion indicator today, and $\hat{Y}_t$ is our baseline prediction for the inversion status 6 months ahead, with t measured in months.

Gradient Boosting Classifier

The Gradient Boosting Classifier (specifically XGBoost) was chosen for its ability to model complex, non-linear feature interactions in high-dimensional data. We tuned the following hyperparameters via Grid Search. `max_depth` controls tree complexity, balancing underfitting vs. overfitting. `learning_rate` scales each boosting step; lower values generalize better when paired with more trees. L1/L2 regularization: L1 performs feature selection by zeroing out weak predictors, while L2 shrinks weights to reduce variance. `scale_pos_weight` upweights the inversion class to counter class imbalance. Finally, we lowered the classification threshold, which shifts the precision/recall tradeoff toward recall, since missed inversions are costlier than false alarms.

Random Forest

Random Forest was chosen for its robustness to overfitting through bagging, and as an independent complement to XGBoost's boosting approach. We tuned `n_estimators`, as more trees reduce variance up to a point. `criterion` sets the splitting rule (e.g. Gini vs. entropy) for measuring node impurity. `max_depth`, similarly to XGBoost, was kept shallow to limit overfitting on the small positive class. `min_samples_leaf` prevents leaves from fitting noise in a few samples. `max_features` limits the number of features considered per split to

decorrelate trees. Finally, `class_weight` reweighs classes to counter imbalance.

Evaluation

Summary Stats

Table 2 reports each model's summary statistics, including accuracy, recall, and precision.

Table 2: Average Performance by Model

Model	Accuracy	Recall	Precision	ROC_AUC	Avg_Precision
Random Forest	0.9000	0.6905	0.5562	0.9129	0.5395
Random Walk	0.9073	0.4405	0.4918	NaN	NaN
XGBoost	0.9049	0.7405	0.5622	0.9268	0.6098

Both machine learning models outperform the Random Walk in recall and precision, while all three models achieve almost identical accuracy. Between the two machine learning models, XGBoost outperforms Random Forest across every metric except accuracy, where Random Forest holds a slight edge.

Visualizations

Figure 2 displays the Precision-Recall and ROC curves for both models. On the ROC curves, Random Forest (AUC = 0.904) and XGBoost (AUC = 0.897) perform similarly, with both curves rising well above the random-chance line, indicating strong overall discrimination between inversion and non-inversion periods. The Precision-Recall curves tell a closer story: Random Forest (AP = 0.431) and XGBoost (AP = 0.428) achieve nearly identical average precision, though both curves show substantial volatility at low recall, reflecting the difficulty of precisely identifying the relatively rare inversion class. Together, these results suggest that while both models are strong classifiers overall (as shown by the high ROC-AUC), predicting

inversions precisely across all recall thresholds remains challenging, a pattern consistent with the class imbalance discussed earlier.

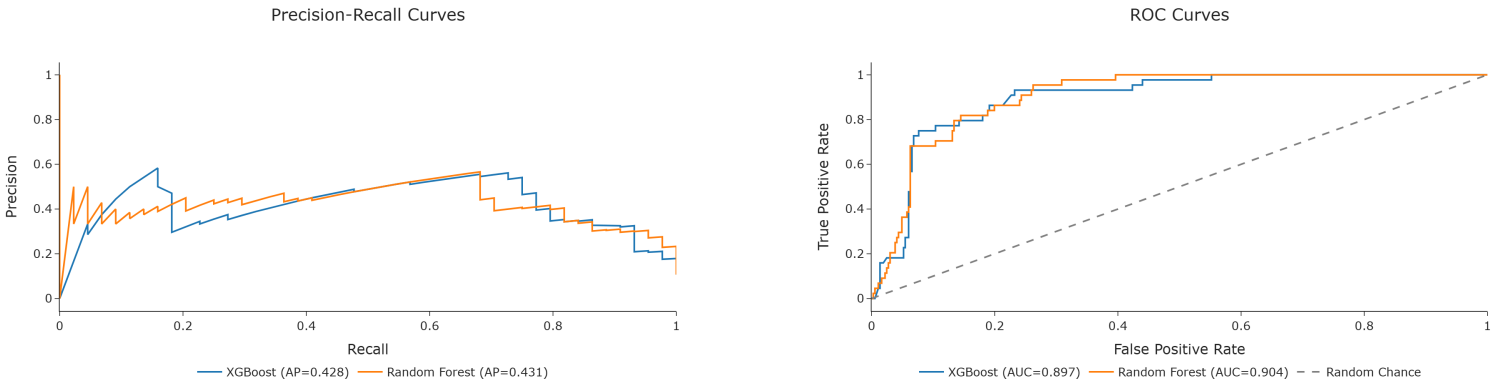


Figure 2: Precision-Recall and ROC Curves

Figure 3 displays the confusion matrices for each model. Compared to the Random Walk baseline, which correctly identifies only 23 of the 44 true inversions, XGBoost correctly identifies 33, a 43.5% increase in correctly detected inversions. Random Forest correctly identifies 30, a 30.4% increase over the baseline. This improvement in detection comes with a tradeoff in false positives: XGBoost produces 28 false positives, a 64.7% increase over the Random Walk's 17, while Random Forest produces 27, a 58.8% increase. XGBoost is the best-performing model overall. It achieves the highest recall of the three models, correctly flagging the largest share of true inversions, and also posts the highest ROC-AUC (0.897) and average precision (0.428) among the two machine learning models. While XGBoost's false-positive count is modestly higher than Random Forest's, this tradeoff is justified given that missed inversions carry far greater economic consequences than false alarms. In a recession-forecasting context, correctly anticipating an inversion that does occur is more valuable than avoiding an occasional false signal, making XGBoost's higher recall the more economically meaningful advantage.

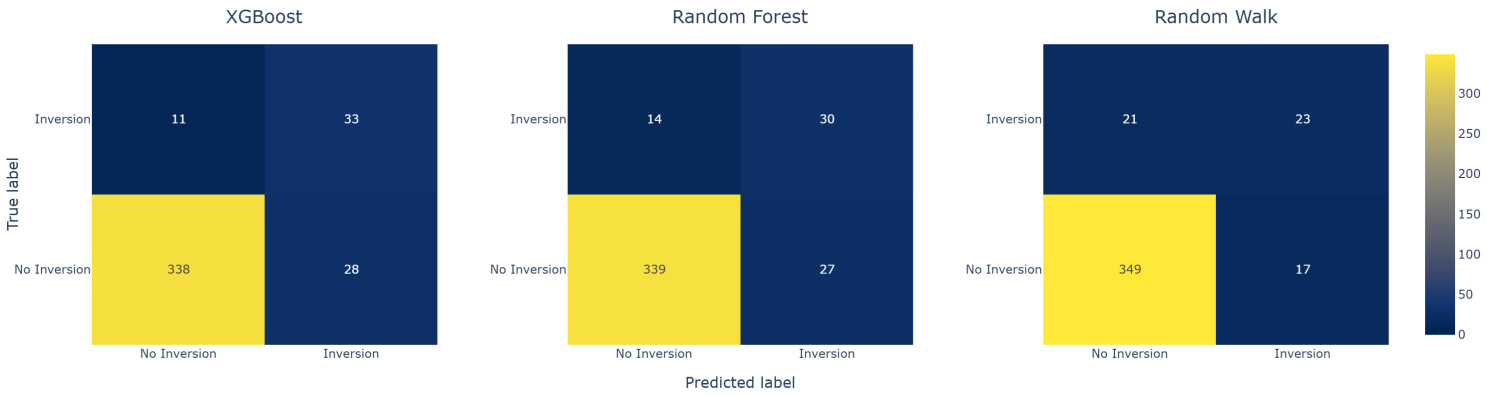


Figure 3: Confusion Matrices

Feature Importance

Figure 4 showcases the feature importance for each model. Both models converge on similar top predictors, most notably AAAFFM and BAAFFM (Aaa and Baa corporate bond spreads over the fed funds rate), along with TSYFFM and HOUST5, suggesting that credit spreads and housing starts carry real predictive power for inversions. The lagged inversion indicator itself carries little importance in either model, despite being the sole driver of the Random Walk baseline, suggesting that macroeconomic indicators offer more forward-looking signal than simple persistence.

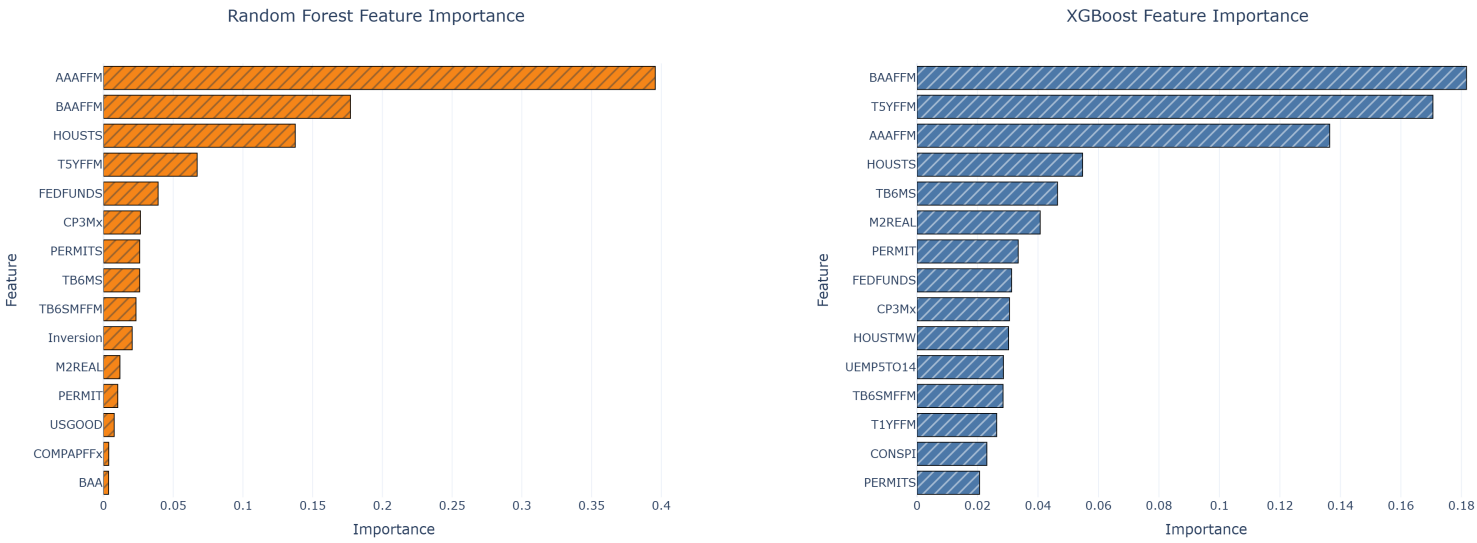


Figure 4: Feature Importance

Conclusion

This report examined whether machine learning can improve prediction of yield curve inversions, and the results offer a measured answer: Machine learning models, particularly XGBoost, do provide a real but modest improvement over a persistence-based baseline, rather than a transformative one. Both XGBoost and Random Forest outperformed the Random Walk baseline in recall, correctly identifying 30-43% more true inversions, but this came with a real tradeoff in false positives and modest average precision below 0.45. XGBoost emerged as the strongest model overall, making it the most economically useful given that missed inversions carry greater costs than false alarms. The feature importance analysis reinforced this modest optimism, as the corporate credit spreads (AAAFFM, BAAFFM) and housing starts were leading indicators, and the lagged inversion status itself carried little predictive weight which suggests that broader macroeconomic variables drive these models edge over the baseline.

Several areas remain for future improvement. With only 44 true inversions in the test set,

model estimates are sensitive to small changes in individual predictions, and techniques such as SMOTE, class-weighted adjustments, or a longer historical sample could improve robustness. The instability observed in the Precision-Recall curves at low recall also points to a need for probability calibration or ensemble smoothing to stabilize predictions.

References

Rayeni, A., & Naderi, H. (2025). Predicting the Canadian Yield Curve Using Machine Learning Techniques. *International Journal of Financial Studies*, 13(3), 170.

Benzoni , L., Chyruk , O., & Kelley, D. (2018). Why Does the Yield-Curve Slope Predict Recessions?. Why does the yield-curve slope predict recessions? - *Federal Reserve Bank of Chicago*

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